

facture, automotive transportation, healthcare and consumer goods. On the basis of global market, digital twin technology is segmented into system twin, parts twin, process twin and product twin. Oracle, General Electric, Microsoft, PTC, ANSYS, Siemens, IBM and Dassault System are some of the leading players in digital twin sector.

NASA was the first to implement digital twin concept. Today, NASA uses digital twins to develop new recommendations, roadmaps, the next-generation vehicles and spacecraft. NASA has used digital twin technology to repair spacecraft travelling outside the range of physical monitoring. For instance, digital twin technology helped NASA engineers and astronauts to repair the ill-fated Apollo 13 mission, bringing the craft and crew back to earth safely.

Future generations of NASA and the US Airforce vehicles will require lighter mass while they are subjected to higher loads and extreme service conditions. NASA spacecraft in future would need to operate for longer periods than the present generation. Conventional approaches used by NASA and the US Airforce have several drawbacks. Digital twin integrates ultra-high fidelity simulation with the vehicles on-board. These are integrated with vehicle health management system, maintenance history and all available historical and flight data to mirror the life of its flying twin, and provide extremely high level of safety and reliability.

Digital twin simulations have been used for development, testing and various other challenges encountered in autonomous vehicle sector. In case of architectural and construction sector, digital twin has arrived as an upgrade of building information modelling.

Digital twin allows analysis of data to identify a potential problem before it occurs, prevents downtime, develops new opportunities and even plans for the future by using simulations. Simply put, digital twin is a bridge between digital and the physical world.

Digital twin helps to create, test and build an equipment in virtual environment. It can be used to train a person to operate a product virtually, even before the actual product is produced, so that the person can learn to handle the product efficiently when it is made available.

Thus the product can be produced

only after ensuring that it would meet the required performance. Today, machine intelligence and connectivity to the cloud allow us a greater potential for large-scale implementation of digital twin technology for companies in a variety of industries.

Improving the performance of the IoT

As more powerful nano sensors are added to devices and processes for improved performance, some standard tool is required to analyse the data collected by the IoT sensors. Digital twin is one way to counter this problem.

The IoT sector is slowly gaining momentum, but it cannot survive without the help of other technologies such as digital twin technology. This technology already has a good track record and has enabled efficient processes that may prove to be better adapters to change and provide efficient services to the IoT sector.

An industrial machinery may have a number of sensors that collect a variety of information like how old it is, how long it has been in use, its production capacity last year/month/week, number of interruptions it had, number of defective units it produced, etc. Such data is sent to the digital twin applications, which process the data, turning it into intelligence and actionable output, so that the defects can be minimised.

Same type of products may face different working conditions. For example, earth moving equipment are subjected to different geographic rock structures at different locations. As such, life span of same-capacity earth moving equipment supplied by a company will be different at different places. Using sensor data, digital twin can easily predict life span of each equipment and can avoid their sudden failure.

Suppose, compressor of an air-conditioning equipment in a shopping mall or office building breaks down. The repair time will be more if this spare part is not available onsite, which can have a huge negative impact on business. Using digital twin technology it is easy to predict such components' failure and send an alert signal to the users or suppliers before such failures. Thus predictive main-

tenance can be done to increase life span and service time of the product.

There will be billions of things represented by digital twin within the next five years. This technology will help companies improve customer experience by enabling better understanding of their needs and develop enhancements to the existing products, operations and services, and even help in further innovation.

General Electric is already using digital wind farms to improve their productivity. This helps them configure each wind turbine prior to its construction. Even the sensor data monitoring wind farm turbines and environment around it can be used to compare with digital twin data. This helps predict failure of a wind turbine before it actually breaks down, so that machine downtime can be considerably reduced.

Prediction is the key feature of the digital twin. For example, digital twin can be used to predict the remaining life of a turbine blade of a specific aircraft engine with great accuracy. This allows the use of condition-based maintenance to manage a specific engine, rather than wasting time with a conventional approach. It also determines the remaining life of the turbine blade after each flight or set of flights, by evaluating operational and environmental data. This way maintenance scheduling becomes more efficient and greatly helps to increase the aircraft engine's life span.

Rolls Royce, Siemens and a few other companies are also using this digital technology since several years now. Tesla uses digital twins to send regular updates about their customers' cars.

Giving new look to digital electronics

Introduced in 2012, digital twin technology has an extremely broad and segmented market that is witnessing a number of exciting trends. With rise of the IoT, digital twins are about to get more attention. According to a report by Gartner, thirteen per cent of organisations have already started using digital twin technology across the globe, and another 62 per cent say they are planning to use it within next twelve months. It is predicted that by 2022, over two-third companies will